“Gamified Environmental Education Platform”

PROJECT OF CSE339

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

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LOVELY PROFESSIONAL UNIVERSITY
PUNJAB

Capstone Project Report

“Gamified Environmental Education Platform”

Technology Stack: MERN (MongoDB, Express.js, React.js, Node.js)

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1. Introduction

1.1 Objective of the Project

The primary objective of this project is to design and develop a full-stack, gamified digital platform that transforms environmental education from a theoretical subject into an engaging, practical, and actionable experience for students in schools and colleges. The platform aims to bridge the gap between knowledge and behavior by incentivizing real-world eco-actions through game mechanics like points, badges, and leaderboards, thereby fostering sustainable habits from a young age.

1.2 Description of the Project

This project is a web-based application built using the MERN stack. It serves as an interactive ecosystem where students can learn about environmental issues, complete real-life eco-tasks (like waste segregation or planting trees), and earn rewards for their efforts. Teachers can monitor participation, verify task completions, and motivate their classes through a dedicated dashboard. The platform leverages gamification to create a competitive yet collaborative environment, encouraging widespread and sustained participation in environmental conservation.

Key capabilities:

- Eco-quizzes and mini-games for concept learning.
- Daily/weekly action-based eco-tasks with evidence upload (photos/text).
- Points, badges, levels, leaderboards and a redeemable eco-merch store.
- Teacher dashboards for verification, analytics and reports.
- Inter-class and inter-school competitions and recognition systems.
- Admin/partner dashboards for sponsorships and analytics.

1.3 Scope of the Project

The scope encompasses the development of a minimum viable product (MVP) with the following core modules:

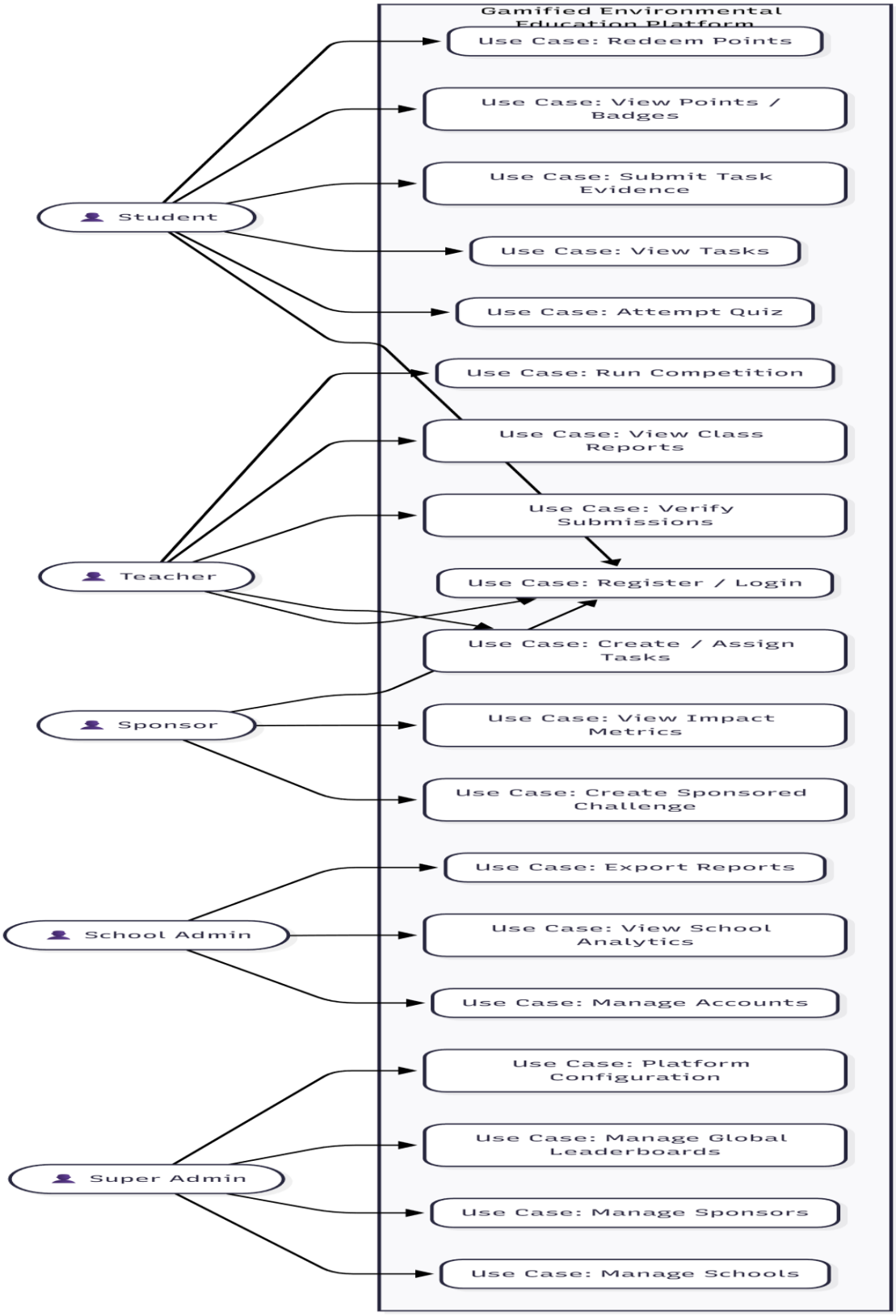
- A Student Portal for accessing quizzes, challenges, and tracking personal progress.
- A Teacher Dashboard for managing classes, verifying tasks, and viewing analytics.
- An Administrative Backend for managing users, content, and the reward system.
- Integration of Gamification Elements including a points system, badges, and leaderboards.

1.3.1 Use Case Model

The primary actors are Students and Teachers.

- Student Use Cases:
    - Register/Log in
    - View and participate in Eco-Quizzes & Mini-Games
    - Accept and submit proof for Daily/Weekly Eco-Challenges
    - View earned Points, Badges, and position on Leaderboards
    - Browse and redeem points in the Eco-Merchandise Store
- Teacher Use Cases:
    - Register/Log in
    - View and manage a list of their students
    - Verify submitted task evidence (photos/checklists)
    - View class/school progress and analytics on a dashboard
    - Generate and view participation reports
- School Admin: Manage Accounts, View School Analytics, Export Reports.
- Super Admin: Manage Schools, Sponsors, Global Leaderboards, Platform Configuration.
- Sponsor: Create Sponsored Challenge, View Impact Metrics. (If you'd like, I can create a UML use-case diagram image next.)

Use Case Diagram



2. System Description

2.1 Customer/User Profiles

- Students (Primary Users): School and college students aged 10 and above. They are motivated by interactive content, recognition, and rewards. They are typically comfortable using smartphones and web applications.
- Teachers/Educators (Secondary Users): Responsible for facilitating the platform's use, verifying student actions, and tracking collective progress. They need a simple, intuitive interface to manage their classes efficiently.
- School Administrators (Tertiary Users): Interested in school-wide analytics, impact reports, and the institution's ranking in inter-school competitions.
- Super Admin: platform-wide configuration, onboarding schools, sponsor management.
- Sponsors / NGOs / CSR teams: create challenges, view impact reports.
- Merchants: provide eco-merch inventory redeemable by points.

2.2 Assumptions and Dependencies

- Assumptions:
    - Users (students and teachers) have access to a smartphone or computer with a reliable internet connection.
    - Schools are willing to integrate the platform into their extracurricular or co-curricular activities.
    - Students will submit genuine evidence for completed tasks.
- Dependencies:
    - Availability of third-party APIs for services like cloud storage (e.g., AWS S3 for image uploads).
    - The project relies on the MERN stack and its associated libraries and frameworks being stable and supported.

2.3 Functional Requirements

Authentication & Profiles

- Student/Teacher/Admin sign-up & login (email / school ID / OAuth).
- Role-based access control.

Learning & Gamification

- Quizzes, mini-games, and learning modules.
- Tasks engine (one-time / recurring tasks).
- Points, badges, levels, progressive unlocks.
- Leaderboards (student/class/school/regional).

Submission & Verification

- Upload photo/text evidence; teacher verification workflow (approve/reject/comment).
- Proof metadata (timestamp, geotag optional).

Rewards & Marketplace

- Points ledger and balance.
- Redeemable eco-merch store with inventory & order management (redeem by points).

Dashboards & Analytics

- Teacher dashboard: student participation, verification queue, class leaderboard.
- School dashboard: school metrics, monthly/term analytics, export reports.
- Super Admin/Sponsor analytics: aggregated impact metrics.

Competitions & Events

- Create time-bound challenges, inter-school leaderboards, award generation.

Notifications & Communication

- Push notifications, email/SMS reminders for tasks/challenges/verification.

Admin & Management

- Manage users, schools, content, sponsor campaigns, merchant accounts.

2.4 Non-Functional Requirements

1. Performance: The application should load pages within 3 seconds and handle concurrent access from multiple users in a school.
2. Usability: The user interface must be intuitive, engaging for students, and easy to navigate for teachers, requiring minimal training.
3. Security: All user data, especially student information, must be protected using encryption, secure authentication, and adherence to data privacy laws.
4. Scalability: The system architecture should be designed to allow for future scaling to accommodate more schools and new features.
5. Availability: The platform should aim for 99% uptime, excluding scheduled maintenance.

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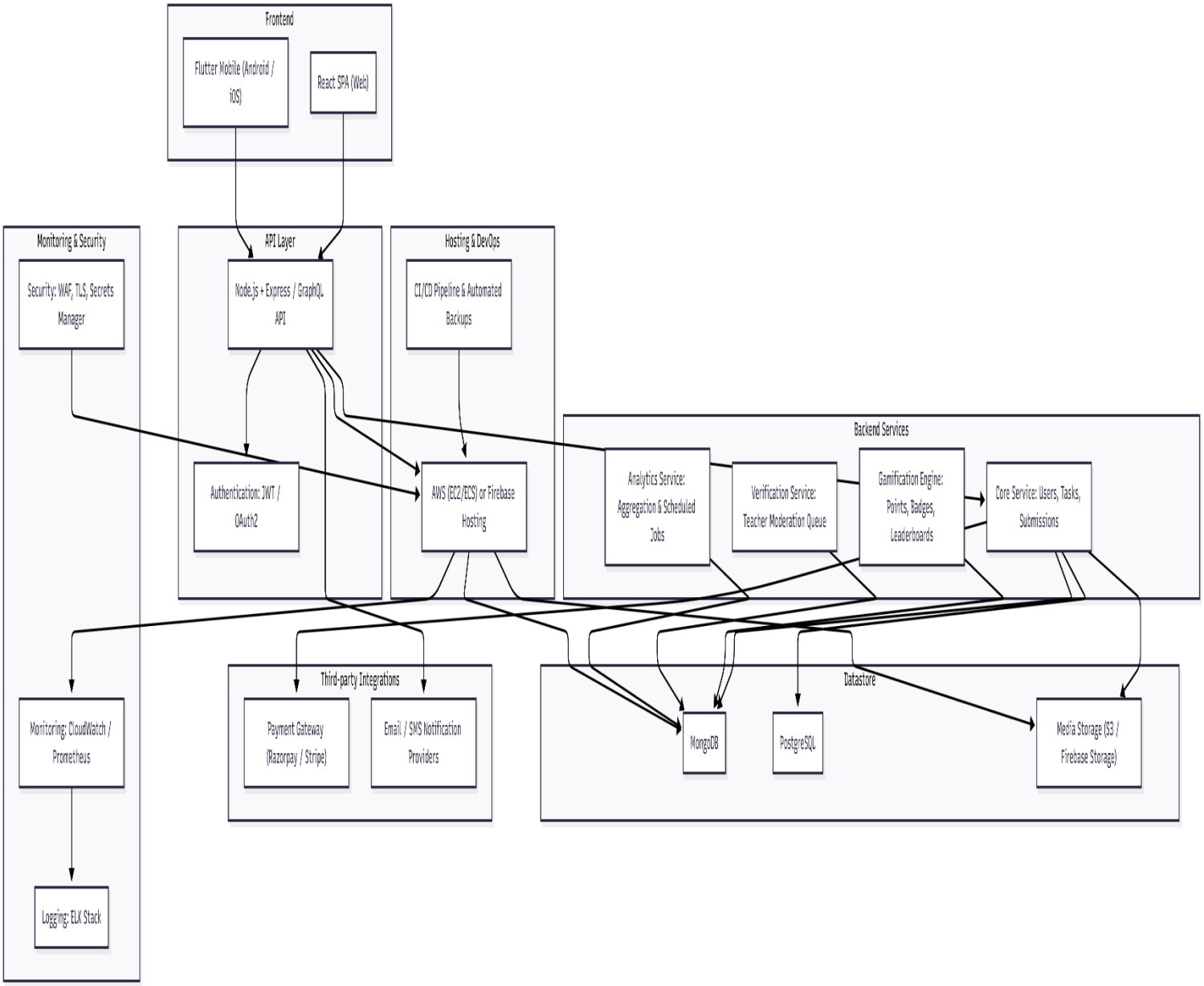
3. Design

3.1 System Design

The system follows a client-server architecture with a React.js frontend and a Node.js/Express.js backend. The backend exposes a RESTful API for all client interactions, and MongoDB serves as the primary database.

1. **Frontend**
    - React.js single-page app for web (student/teacher/admin interfaces).
    - Flutter mobile app (Android/iOS).
2. **API Layer**
    - RESTful endpoints (or GraphQL optional) built with Node.js + Express (or Django).
    - Authentication & Authorization (JWT/OAuth).
3. **Backend Services**
    - **Core service**: user management, tasks, submissions.
    - **Gamification engine**: points calculation, badges, leaderboards.
    - **Verification service**: teacher moderation queue.
    - **Analytics service**: aggregates metrics, scheduled jobs for reports.
4. **Datastore**
    - Primary DB: MongoDB (for flexible task/submission data) or PostgreSQL (if relational integrity preferred).
    - Media storage: S3 (or Firebase Storage).

5. **Third-party integrations**
    - Payment gateway for paid features/merch (Razorpay/Stripe).
    - Email/SMS providers for notifications.
6. **Hosting & DevOps**
    - AWS (EC2/ECS, RDS/DocumentDB, S3), or Firebase (Auth, Firestore, Storage) for quicker MVP.
7. **Monitoring & Security**
    - Logging (ELK), monitoring (CloudWatch/Prometheus), automated backups.



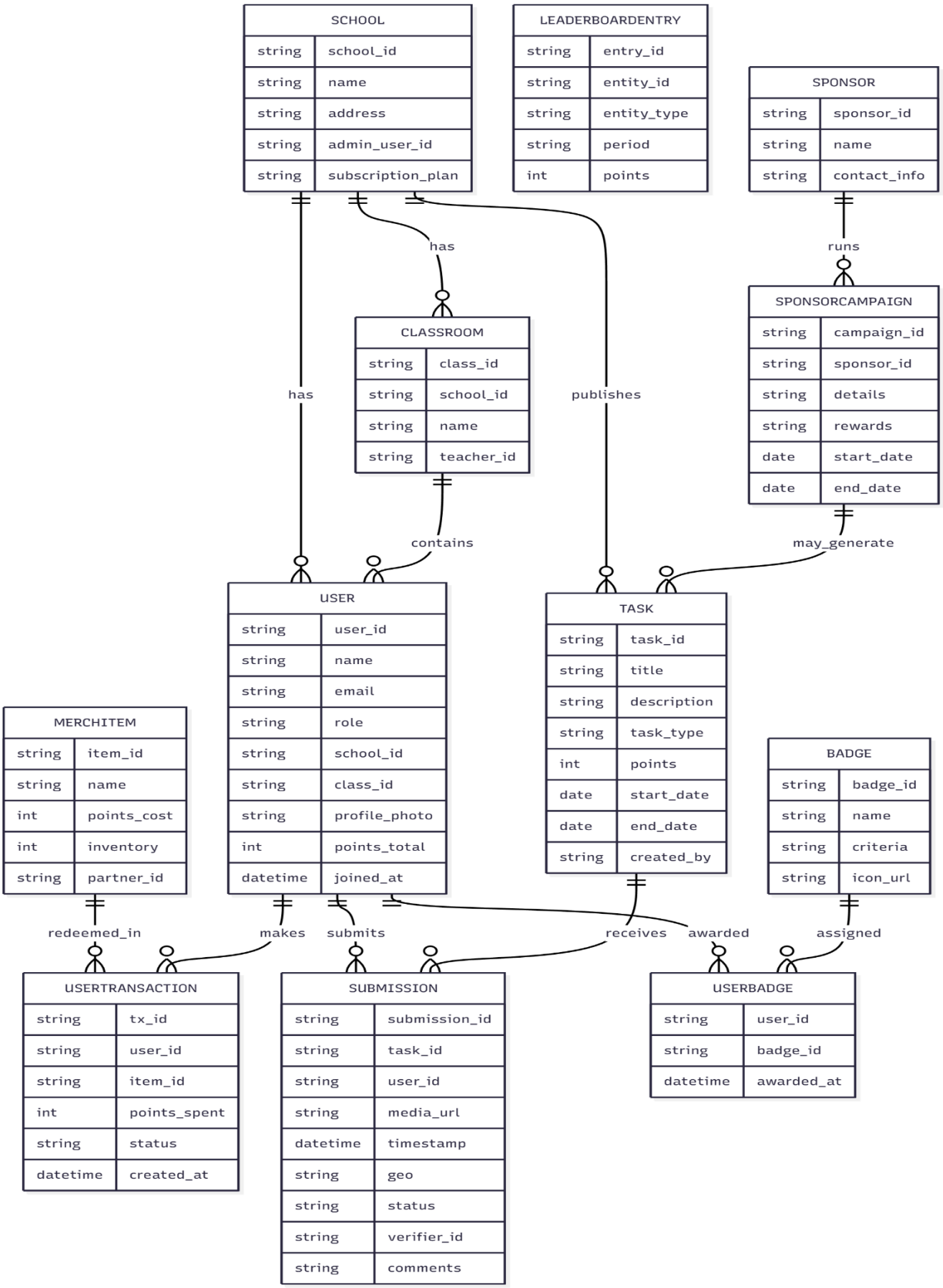
3.1.1 E-R Diagram

Entities:

- **User** (user_id, name, email, role [student/teacher/admin], school_id, class_id, profile_photo, points_total, joined_at)
- **School** (school_id, name, address, admin_user_id, subscription_plan)
- **Class** (class_id, school_id, name, teacher_id)
- **Task** (task_id, title, description, task_type, points, start_date, end_date, created_by)
- **Submission** (submission_id, task_id, user_id, media_url, timestamp, geo, status [pending/approved/rejected], verifier_id, comments)
- **Badge** (badge_id, name, criteria, icon_url)
- **UserBadge** (user_id, badge_id, awarded_at)
- **LeaderboardEntry** (entity_id, entity_type [student/class/school], period, points)
- **MerchItem** (item_id, name, points_cost, inventory, partner_id)
- **Transaction** (tx_id, user_id, item_id, points_spent, status, created_at)
- **SponsorCampaign** (campaign_id, sponsor_id, details, rewards, start, end)

Key Relationships:

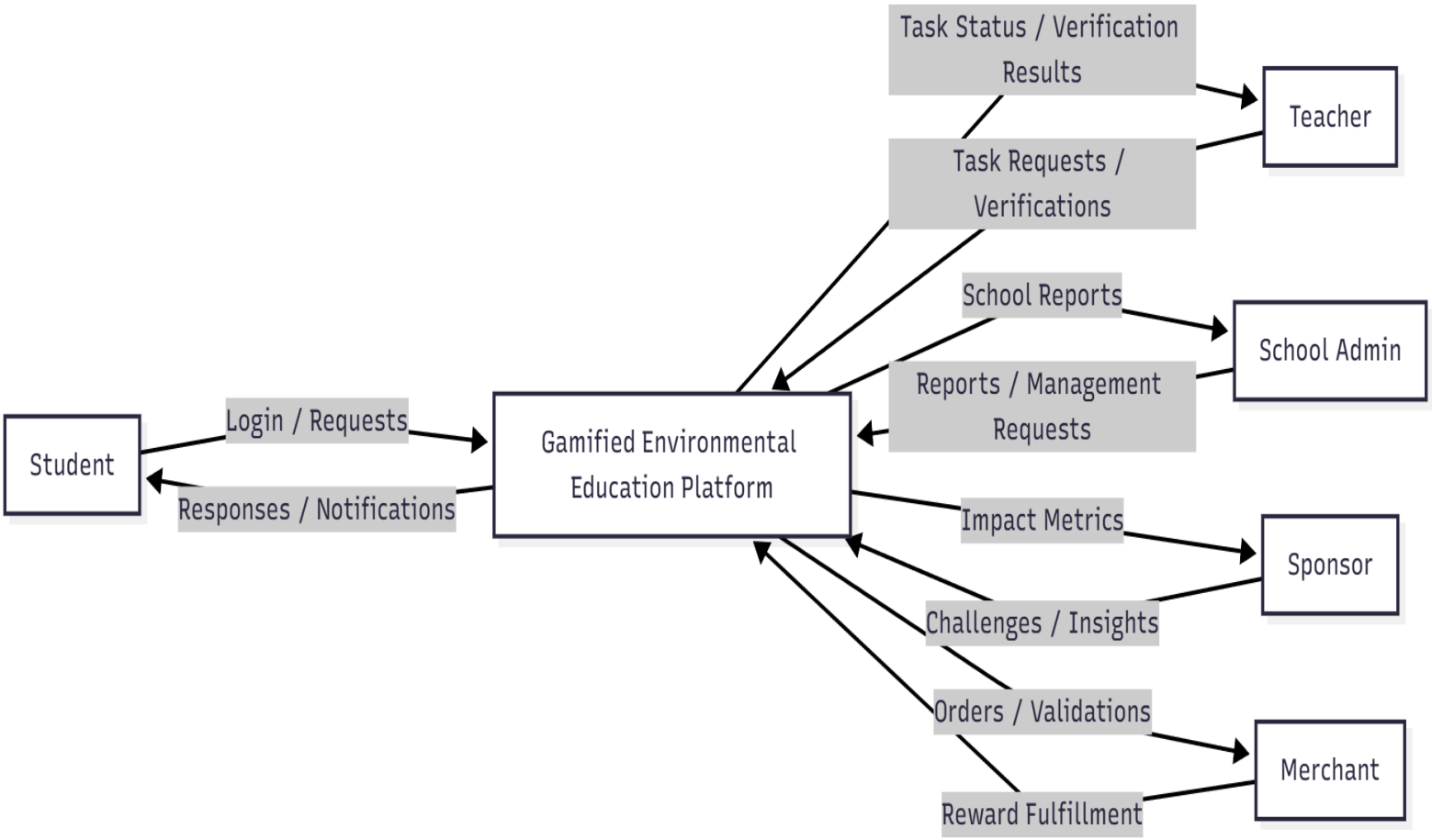
- User ⇄ School (many-to-one)
- User ⇄ Class (many-to-one)
- Class ⇄ School (many-to-one)
- Task ⇄ School (tasks may be school-level or global)
- Submission ⇄ Task (many submissions for a task)
- Submission ⇄ User (each submission by one user)
- User ⇄ UserBadge (many-to-many via UserBadge)
- User ⇄ Transaction (one-to-many)


SCHOOL
string school_id
string name
string address
string admin_user_id
string subscription_plan
LEADERBOARDENTRY
string entry_id
string entity_id
string entity_type
string period
int points
SPONSOR
string sponsor_id
string name
string contact_info
has
runs
CLASSROOM
string class_id
string school_id
string name
string teacher_id
has
publishes
SPONSORCAMPAIGN
string campaign_id
string sponsor_id
string details
string rewards
date start_date
date end_date
contains
may_generate
USER
string user_id
string name
string email
string role
string school_id
string class_id
string profile_photo
int points_total
datetime joined_at
TASK
string task_id
string title
string description
string task_type
int points
date start_date
date end_date
string created_by
MERCHITEM
string item_id
string name
int points_cost
int inventory
string partner_id
BADGE
string badge_id
string name
string criteria
string icon_url
redeemed_in
makes
submits
receives
awarded
assigned
USERTRANSACTION
string tx_id
string user_id
string item_id
int points_spent
string status
datetime created_at
SUBMISSION
string submission_id
string task_id
string user_id
string media_url
datetime timestamp
string geo
string status
string verifier_id
string comments
USERBADGE
string user_id
string badge_id
datetime awarded_at

3.1.2 DFD's (Data Flow Diagrams)

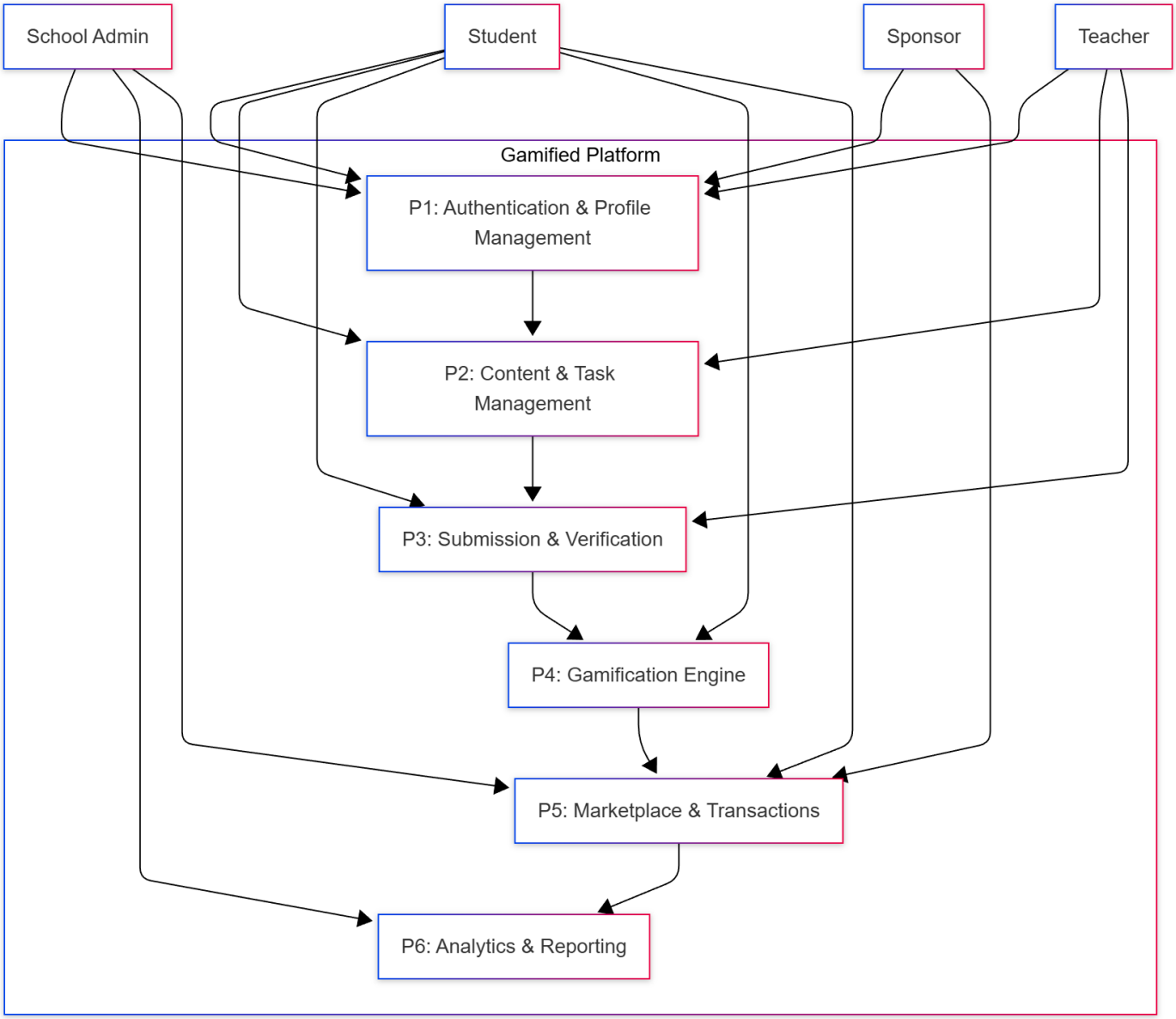
Context diagram (DFD level 0)

- External actors: Student, Teacher, School Admin, Sponsor, Merchant.
- System: Gamified Platform (single process).
- Flows: Login/Requests, Task Submissions, Verifications, Rewards Redemptions, Analytics Reports.



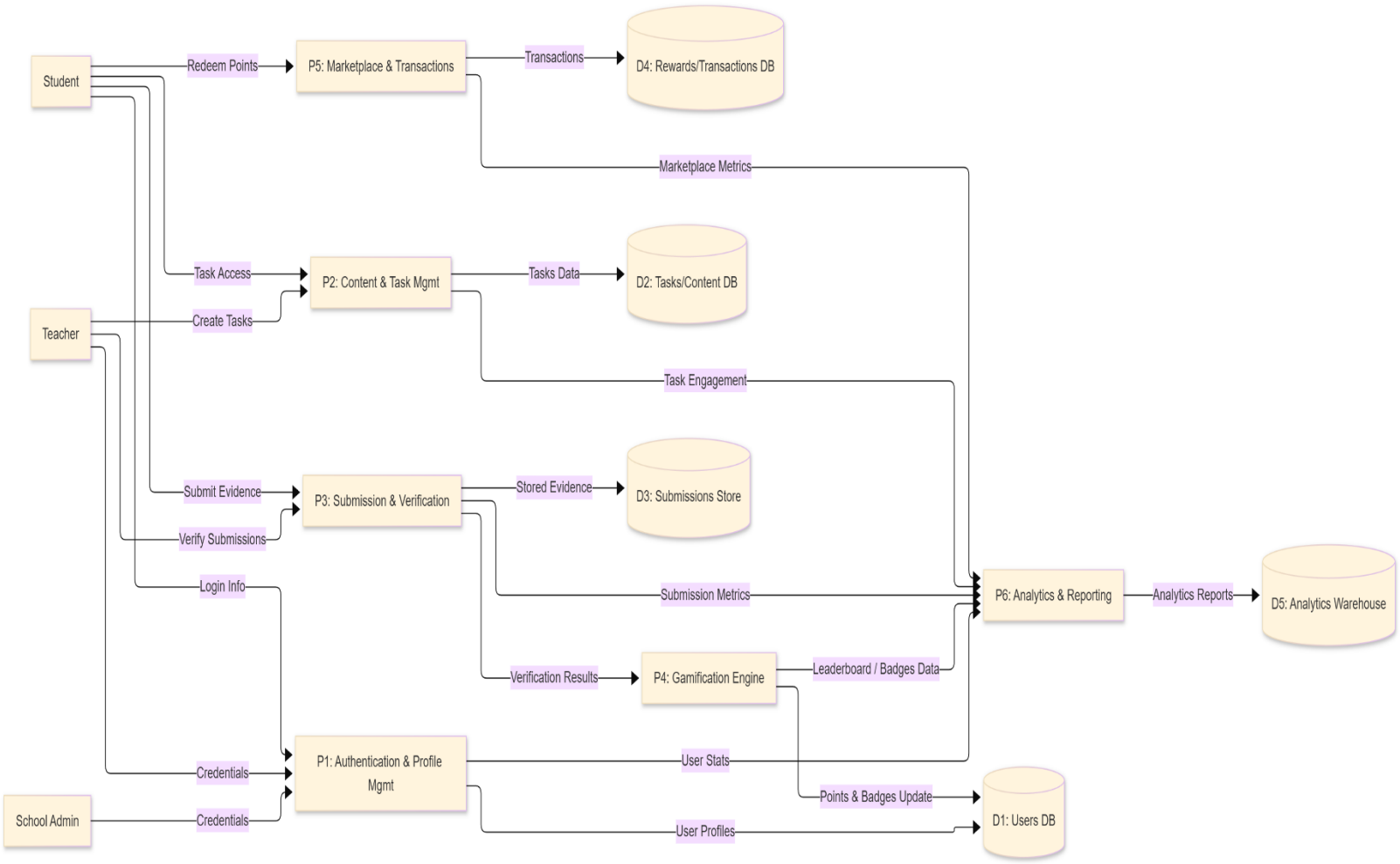
DFD Level 1 — main processes

- P1: Authentication & Profile Management
- P2: Content & Task Management (create tasks, quizzes)
- P3: Submission & Verification (upload → teacher verify)
- P4: Gamification Engine (points, badges, leaderboards)
- P5: Marketplace & Transactions
- P6: Analytics & Reporting



Data stores

- D1: Users DB
- D2: Tasks/Content DB
- D3: Submissions (media store + metadata)
- D4: Rewards/Transactions DB
- D5: Analytics Warehouse



3.2 Database Design

The database will be implemented in MongoDB, a NoSQL database, chosen for its flexibility in storing varied data like user submissions and challenge logs.

Sample Collections (equivalent to tables in SQL):

- **users**

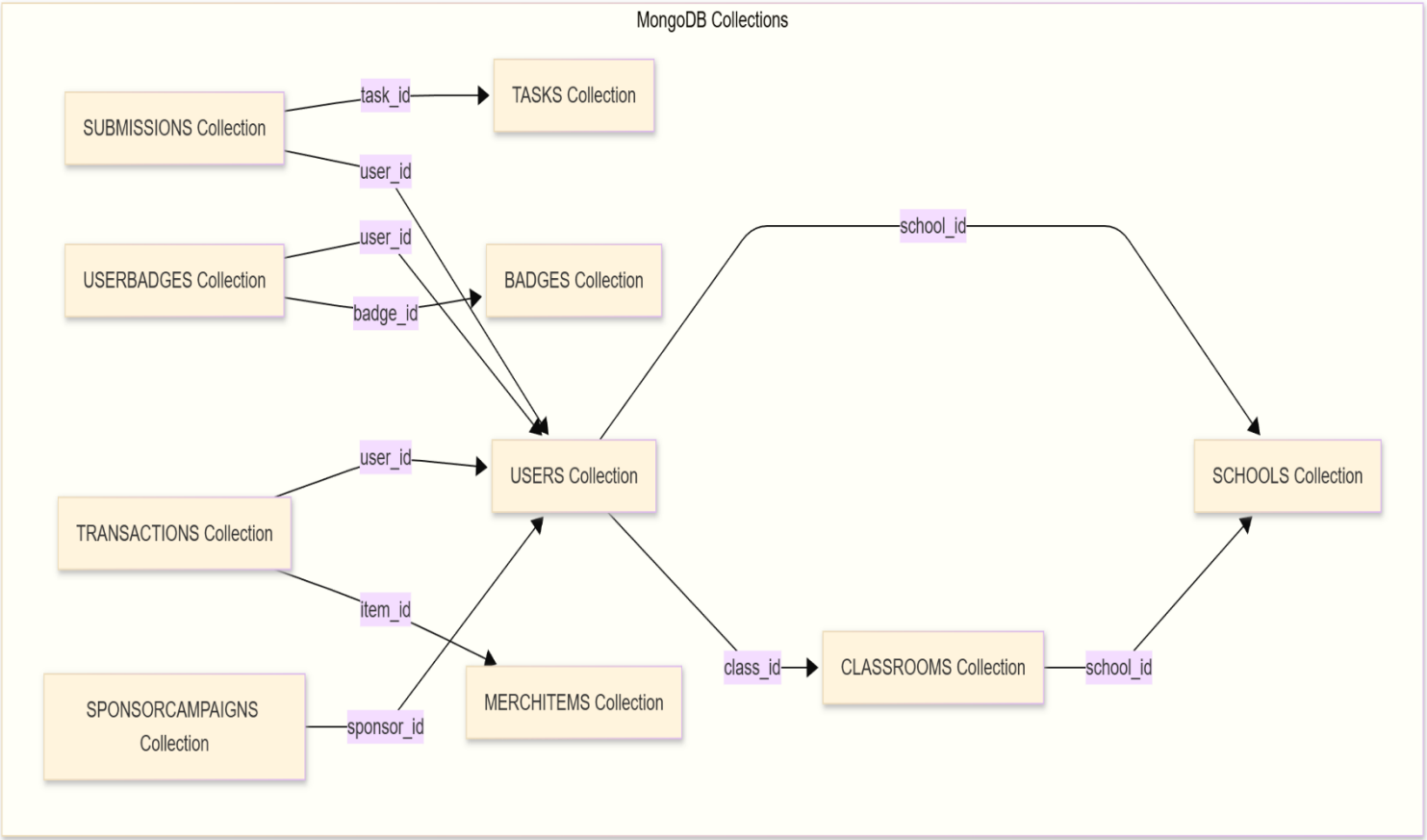
```
{
  "_id":"ObjectId",
  "name":"Ram Kumar",
  "email":"ram@school.edu",
  "role":"student",
  "school_id":"ObjectId",
  "class_id":"ObjectId",
  "points_total": 1250,
  "badges": ["green_warrior_id"],
  "created_at": "ISODate"
}
```

- **tasks**

```
{
  "_id":"ObjectId",
  "title":"Plastic-Free Week",
  "description":"Avoid single-use plastic for 7 days",
  "points": 100,
  "type":"challenge",
  "start_date":"ISODate",
  "end_date":"ISODate",
  "created_by":"teacher_id"
}
```

- **submissions**

```
{
  "_id":"ObjectId",
  "task_id":"ObjectId",
  "user_id":"ObjectId",
  "media":[ "s3://...jpg" ],
  "timestamp":"ISODate",
  "status":"pending",
  "verifier_id":null
}
```



4. Scheduling and Estimates

This schedule outlines a 6-month development timeline for the MVP(Minimum Viable Product).

Phase	Duration	Key Deliverables
Phase 1: Planning & Design	Month 1	Finalized SRS, UI/UX Wireframes, System Architecture, ER Diagram
Phase 2: Core Backend Development	Months 2-3	Set up Node.js/Express server, MongoDB models, User Authentication & Authorization APIs, Challenge & Submission CRUD APIs
Phase 3: Frontend Development	Months 3-4	React.js components for Login, Student Dashboard (challenges, leaderboard), Teacher Dashboard (verification queue)
Phase 4: Gamification & Integration	Month 5	Implement points system, badge awarding logic, real-time leaderboard updates, integrate image upload feature
Phase 5: Testing & Deployment	Month 6	Unit & Integration Testing, User Acceptance Testing (UAT) with a pilot school, Deploy application on a cloud service (e.g., Heroku/AWS)

Technology Estimates:

- Frontend: React.js, HTML5, CSS3, JavaScript (ES6+)
- Backend: Node.js, Express.js
- Database: MongoDB with Mongoose ODM
- Authentication: JWT (JSON Web Tokens)
- File Storage: AWS S3 or similar cloud service
- Version Control: Git/GitHub
- Deployment: Frontend on Netlify/Vercel, Backend on Heroku/Railway.

5. Literature Survey – Gamified Environmental Education Platform

Environmental education has gained significant importance in recent years due to rapid climate change, biodiversity loss, and the increasing need for sustainable lifestyles. Traditional classroom-based teaching methods alone are not sufficient to inspire behavioral change among students. With advancements in digital technologies, gamification, e-learning, and mobile platforms, new approaches are emerging to enhance environmental awareness and drive active participation. This literature survey presents existing research, theories, and systems relevant to the development of a gamified environmental education platform.

5.1. Gamification and Learning Engagement

Deterding et al. (2011) introduced gamification as the application of game design elements—such as points, leaderboards, levels, and badges—in non-game contexts to enhance user motivation and engagement. Their work forms the conceptual foundation of modern gamified educational systems. Hamari, Koivisto, and Sarsa (2014), through an extensive literature review, confirmed that gamification positively impacts learning participation, although effectiveness depends greatly on thoughtful design and the learner's internal motivations.

Werbach and Hunter (2012) highlight that a successful gamification framework must include meaningful rewards, clear progress indicators, and achievable challenges. These principles directly support the design of the platform's points system, badge mechanics, eco-tasks, and leaderboard competitions.

5.2. Motivation Theory in Educational Systems

Motivation plays a crucial role in behavioral change and learning efficiency. Deci and Ryan's Self-Determination Theory (1985) emphasizes that intrinsic motivation grows when autonomy, competence, and relatedness are supported. These psychological needs can be effectively addressed through gamified activities, progress tracking, social comparison, and peer interactions.

Similarly, Bandura's Social Learning Theory (1977) suggests that individuals learn behaviors through observation, imitation, and reinforcement. In the context of this project, leaderboards, peer achievements, and group-based eco-challenges work as social motivators that encourage students to adopt environmentally responsible habits.

5.3. Environmental Education and Sustainable Development Goals (SDGs)

UNESCO's framework for Education for Sustainable Development (2019) stresses experiential and participatory learning approaches. Students learn better when they engage in real-world actions — such as recycling, reducing waste, or saving electricity — rather than merely reading about environmental issues. The proposed platform aligns strongly with this by allowing students to complete eco-tasks and upload evidence for teacher verification.

Studies by Palmer (2016) and Jensen (2002) further argue that environmental education should be action-oriented and community-based. Gamified environmental platforms bridge the gap between theory and practice by making sustainability fun, measurable, and interactive.

5.4. Digital & Mobile Learning Platforms

The integration of digital platforms in education has transformed how content is created, shared, and consumed. Goyal et al. (2021) found that mobile learning significantly enhances accessibility, convenience, and engagement—especially for younger learners. This supports the inclusion of a Flutter-based mobile app for Android/iOS alongside the React web interface.

Digital platforms equipped with micro-learning modules, quizzes, and interactive visuals have been shown to improve retention rates (Chiang & Lee, 2019). These findings justify the platform's inclusion of gamified quizzes, eco-concepts, and personalized dashboards.

5.5. Existing Environmental Apps & Their Limitations

Several mobile applications and web-based platforms aim to promote eco-friendly behavior. Apps like **JouleBug**, **Oroeco**, and **Greenly** encourage sustainable habits through streaks, points, and carbon tracking. However, most of these platforms primarily target adults and lack:

- Structured school/college integration
- Teacher verification workflows
- Classroom-level competitions
- Curriculum-aligned environmental tasks
- Institutional analytics and reporting

Research by Karatas et al. (2022) shows that environmental gamification works best when integrated into structured educational environments rather than as standalone lifestyle apps. This highlights the need for a school-centric platform like yours.

5.6. Data Management & Technology Foundations

The use of NoSQL databases like MongoDB for scalable educational systems is supported by MongoDB Inc. (2023), which states that flexible schema design and horizontal scalability make NoSQL ideal for handling dynamic content such as student submissions, media files, and gamification logs.

Studies in cloud computing (Armbrust et al., 2010) also highlight the importance of distributed systems, elasticity, and high availability — which validate the choice of AWS/Firebase for hosting the platform.

Additionally, MERN stack applications (MongoDB, Express.js, React.js, Node.js) are widely used in educational and enterprise systems due to:

- Fast development cycles
- JSON-based data handling
- Efficient real-time updates
- Scalability for large student populations

5.7. Gaps Identified in Existing Literature & Systems

Despite significant work in gamification and environmental education, certain gaps remain:

Gap Identified	**How the Proposed Project Addresses It**
Lack of school-integrated gamified systems	Platform includes students, teachers, school admins, and sponsors.
Limited real-world eco-action verification	Teacher approval workflow with media evidence.
Poor alignment with curriculum	Tasks, quizzes, and challenges tailored for educational use.
Weak analytics	Built-in dashboards for teachers, admins, sponsors.
No reward redemption ecosystems	Marketplace with eco-merch redeemable through points.
Lack of multi-role system support	Platform supports students, teachers, admins, sponsors, merchants.

Our platform fills these gaps by combining **gamification + environmental action + academic structure**, making it more comprehensive than existing solutions.

5.8. Summary of the Literature

The reviewed literature reveals that:

- Gamification significantly boosts learner motivation and engagement.
- Psychological theories support the use of rewards, social comparison, and reinforcement.
- Environmental education is most effective when experiential, action-based, and measurable.
- Mobile and web platforms enhance accessibility and participation.
- Current eco-apps lack institutional integration, analytics, and curriculum support.
- MERN stack and NoSQL databases provide an efficient foundation for scalable and interactive systems.

Hence, the proposed **Gamified Environmental Education Platform** fills a crucial gap by integrating environmental education, teacher workflows, gamification, and institutional analytics into one unified system.